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CS-499

Prof Ogoh

Milestone 4 Narrative

The artifact that I selected for this narrative is the Grazioso Salvare Animal Shelter application that I initially developed during CS-340. The application utilizes a MongoDB database and Python to manage animal shelter records and allowing users to perform create, read, update, and deletion operations. The system also allows the users to filter data to identify animals that meet the specific criteria for rescue training programs.

I selected this artifact because it directly demonstrates database integration within an application and provides a strong foundation for expanding database related functionality. While the original implementation allowed for basic CRUD operations it does lack more advanced querying capabilities that would be expected within a professional database driven application. The enhancement focuses on improving the database interactions through advanced query functions such as refined filters, parameterized queries, and improved data validation before database operations are executed. I have also enhanced the structure of the database access by ensuring consistent handling of queries and improving how the results are processed and returned. These improvements increase the efficiency and reliability of the application while aligning it with industry best practices for database management.

The improved query structure and validation demonstrate my ability to design and evaluate computing solutions using the appropriate database practices. The implementation of structured and efficient database interactions reflects the use of solid techniques and tools to

deliver reliable results. Additionally, by incorporating validation and controlled query handling I have demonstrated a security mindset by reducing potential risk of invalid data and unintended database operations.

Throughout this enhancement I developed a deeper understanding of how database design and query structure impact application performance and reliability. One challenge I encountered during my enhancement was ensuring that the enhanced queries remained compatible with the existing logic in the application while improving overall efficiency. Addressing this required careful testing to ensure that the CRUD operations continued to function after the query changes.